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S&P volatility, VIX, and asymptotic volatility estimates

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ABSTRACT

We examine the efficacy of the VIX as a predictor of 30-day forward S&P 500 volatility. We find that its accuracy hovers between 20% and 25%, depending on sampling period. An alternative framework, built on asymptotic distribution theory (AVE), predicts this volatility with an accuracy between 90% to 95%. The adjusted R-square adjudicates for accuracy. Other goodness-of-fit measures corroborate this evidence. We suggest that this outcome is underpinned by the fact that (a) options price behaviors do not adequately reflect stock market volatility patterns, and (b) our methodology accounts more comprehensively for idiosyncratic risk.

1. Introduction

Financial market volatility has been extensively studied in different forms and approaches (e.g., Patton (2011), Hansen and Lunde, A. (2005), Liu et al., (2022); Wang et al., (2022), and Lyócsa and Plíhal (2022), among others). Our focus is on providing an alternative to the CBOE volatility index, VIX, as a predictor of S&P 30-day forward volatility.

The VIX, formulated by the Chicago Board of Options Exchange (CBOE) in 1993 “to measure the market expectation of 30-day volatility” (CBOE VIX Whitepaper (2019)) is widely employed by market participants. Originally based on the Black-Scholes implied volatility of at-the-money options on the S&P100, the VIX was redesigned in 2003 and later in 2014 to be model-free and no longer relies on an option pricing framework. Rather, it depends on an amalgam of price quotations of at- and out-of-the-money puts and calls for the S&P 500 index (CBOE (2019)).¹ The VIX is intended to represent the square root of the strike price on a zero-value variance swap, with payoff = $(\sigma_R^2 - K)$, where σ_R^2 is the variance of the realized returns over the next 30-day period and K is the strike price equal to $\sqrt{E(\sigma_{R,t,t+30}^2)}$ (see Demeterfi et al., (1999)). Not surprisingly, the index has become a benchmark for expected stock market volatility (CBOE (2019), Whaley (2009)).

Even though the modern VIX formulation is option-model free, it still relies on the distributional assumptions of underlying returns (Geometric Brownian Motion), and ultimately on the market prices of the options themselves (e.g., Jiang and Tian (2005); Carr and Wu (2009), Ait-Sahala et al. (2015), among others). Importantly, if returns have jumps, the approximation of K will have errors that may be substantial, persistent, and not centered at zero. In fact, the VIX is known to systematically overestimate future realized volatility, representing persisting losses to variance swap holders (e.g., Bakshi and Kapadia (2004) and Bekaert and Hoerova (2014)).

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¹ “There were two changes to the computation of the VIX; one in 2003 and the other in 2014. See, for example, <https://quant.stackexchange.com/questions/25157/how-was-the-old-vix-calculated>.

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But beyond this variance-premium bias, the VIX is widely known to be asymmetrically associated with current volatility – the intensity of its rise, on average, in falling markets is much stronger, than the intensity of its fall in rising markets, possibly due to hedging pressures (e.g., [Bollen, and Whaley \(2004\)](#), [Whaley 2009](#)). That is, the VIX appears to better indicate investors' collective fear than their collective aggression. Model-free it might be, the VIX still relies on option prices that may be driven as much by current *returns* as by expectations of *volatility*. This has led to suggestions that the VIX does not really measure return volatility, but rather a combination of return moments ([Chow et al., \(2020\)](#)).

To what extent does the VIX really project future volatility? Is it possible to provide a better indicator of future S&P volatility that altogether does not rely on options quotations? Would a simpler-to-implement standard statistical framework that captures the essence of return patterns (e.g., autoregression/serial correlation with innovations) serve volatility forecasting better? To address the question, we first regress the 30-day S&P realized volatility on the VIX. Given a date (say 9/20), we calculate the return of the S&P for the next 30 days and use the standard deviation of these returns as the realized volatility. From the VIX regression, we use the adjusted R-square as a proxy for the model fitness, and find that it hovers around 20% to 25%, depending on sampling period. We employ several samples, a "full" sample from 1990 to 2021, sub-samples from 1990 to 2002, from 2004 to 2013, and from 2015 to 2021 to account for changes in VIX computation.

Next, we propose an alternative framework to calculate future realized S&P volatility. Our approach utilizes a key feature of distribution theory and inferences from the seminal work of [Tauchen \(1985\)](#) to develop a metric called the Asymptotic Volatility Estimator (AVE). We then regress 30-day future S&P realized volatility on the AVE, and find that the adjusted R-square in all samples is in excess of 90%. Given that the use of volatility proxies must be looked at with caution (see [Patton \(2011\)](#)), we corroborate the evidence (from the adjusted R-square) with evidence from other goodness-of-fit metrics. The overall evidence is consistent and overwhelming – our approach outperforms the VIX index.

2. Empirical framework

Our theoretical framework which is based on asymptotic distribution theory may be viewed as a classic time-series forecasting model based on autoregression, serial correlation and an innovation term. [Tauchen \(1985\)](#) presents the computational basis on how to use an AR(1) model to discretize two or more states within the Markov chain framework. This framework and its extensions are standard in the literature (e.g., [Bonaparte, Cooper, and Zhu \(2012\)](#)). For our purpose, the forecasting is based on S&P returns volatility itself, and not on options prices.

We introduce the S&P return process as an autoregressive process of order 1, denoted by AR (1). Letting R_t denote the return of the S&P at time t , the AR (1) process is as follows:

$$R_t = c + \rho * R_{t-1} + \varepsilon_t \quad (1)$$

where c is a constant, ρ is the serial correlation parameter, and ε_t is the error term which reflects unobservables. The error term is assumed to be normally distributed with 0 mean and standard deviation σ_ε^2 , hence, $\varepsilon \sim N(0, \sigma_\varepsilon^2)$; the quantity σ_ε^2 is the innovation term.

To recast the notation into a rotating time-series window, we present [Eq. \(1\)](#) in a slightly different form, where R_t denotes the S&P return for given window j at time t so [Eq. \(1\)](#) then becomes:

$$R_{j,t} = c_{j,t} + \rho_{j,t} * R_{j,t-1} + \varepsilon_{j,t} \quad (2)$$

Next, we present our approach for quantifying AVE. Let the distribution function of ε_{jt} be independent and identically distributed (iid) where F is the cumulative distribution with unit variance. Let $R_{j,t}$ denote the discrete-valued process that approximates the continuous-valued process and $R_{j,t}^{S\&P-1} < R_{j,t}^{S\&P-2} \dots < R_{j,t}^{S\&P-N}$. A method for selecting the values $\bar{R}_{j,t}^{S\&P-1}$ is to let $\bar{R}_{j,t}^{S\&P-N}$ be a multiple m of the unconditional standard deviation $\sigma_y = \left(\frac{\sigma_{\varepsilon}^2}{1-\rho_{\varepsilon}^2} \right)^{0.5}$. Then let $\bar{R}_{j,t}^{S\&P-1} = -\bar{R}_{j,t}^{S\&P-N}$, and let the remaining be equispaced over the interval $[\bar{R}_{j,t}^{S\&P-k}, \bar{R}_{j,t}^{S\&P-N}]$. Denoting $w = \bar{R}_{j,t}^{S\&P-k} - \bar{R}_{j,t}^{S\&P-k-1}$, the method for calculating the transition matrix is

$$P_{ik} = \Pr \left[\bar{R}_{j,t}^{S\&P-k} - \frac{w}{2} \leq \bar{R}_{j,t}^{S\&P-k} + \varepsilon_t \leq \bar{R}_{j,t}^{S\&P-k} - \frac{w}{2} \right]$$

$$P_{ik} = F \left(\frac{\bar{R}_{j,t}^{S\&P-k} - \rho \bar{R}_{j,t}^{S\&P-i} + w/2}{\sigma_\varepsilon} \right) - F \left(\frac{\bar{R}_{j,t}^{S\&P-k} - \rho \bar{R}_{j,t}^{S\&P-i} - w/2}{\sigma_\varepsilon} \right)$$

Then,

$$P_{i1} = F \left(\frac{\bar{R}_{j,t}^{S\&P-1} - \rho \bar{R}_{j,t}^{S\&P-i} + w/2}{\sigma_\varepsilon} \right)$$

$$\text{and } P_{iN} = F \left(\frac{\bar{R}_{j,t}^{S\&P-N} - \rho \bar{R}_{j,t}^{S\&P-i} - w/2}{\sigma_\varepsilon} \right)$$

Therefore, our policy function for the AVE_t is:

$$AVE_t = f(\rho_t, \sigma_{t,\varepsilon}^2) \quad (3)$$

We now specify the policy function $f(\rho_t, \sigma_{t,\varepsilon}^2)$. From the parameters ρ_t and $\sigma_{t,\varepsilon}^2$ we create two variables, $\tau_{jt,1}$ and $\tau_{jt,2}$. Specifically, $\tau_{jt,1}$ measures the probability for turning to a high state and $\tau_{jt,2}$ represents the return in the high state, and we assume a two-state model.

$$\tau_{jt,1} = P_{iH} = F\left(\frac{\bar{R}_{jt}^{S\&P-1} - \rho \bar{R}_{jt}^{S\&P-i} + w/2}{\sigma_\varepsilon}\right)$$

$$\tau_{jt,2} = \bar{R}_{jt+1}^{S\&P-H} = \left(\frac{\sigma_{t,\varepsilon}^2}{1 - \rho_t^2}\right)^{0.5}$$

Given this parametrization, the next step is to estimate the policy function for AVE. If we represent $Var(R_{jt+1}^{S\&P})$ as the future 30-day realized return volatility, and regress this following model:

$$Var(R_{jt+1}^{S\&P}) = Constant_{jt} + \phi_{1j} * \tau_{jt,1} + \phi_{2j} * \tau_{jt,2} + \phi_{3j} * \rho_{jt} + \phi_{4j} * \sigma_{t,\varepsilon}^2 + \vartheta_{jt} \quad (4)$$

then ϕ_{1j} , ϕ_{2j} , ϕ_{3j} , and ϕ_{4j} are parameters to estimate, and ϑ_{jt} is the error term.

Given the estimates for the four parameters, AVE is then calculated as:

$$AVE = Constant_{jt} + \phi_{1j} * \tau_{jt,1} + \phi_{2j} * \tau_{jt,2} + \phi_{3j} * \rho_{jt} + \phi_{4j} * \sigma_{t,\varepsilon}^2 \quad (5)$$

To perform our analyses, we collect daily data from CRSP about the S&P index for the period 1990–2021.

3. Empirical analyses

In this section we examine the performance of the VIX and the AVE. Our analysis is based on the regression

$$Var(R_{t+1}^{S\&P}) = c + \beta * AVE_t + \varepsilon_t$$

Here $Var(R_{t+1}^{S\&P})$ is the 30-day forward volatility from time t and AVE_t is the Asymptotic Volatility estimator at time t . The constant is denoted by c and β is the co-movement between AVE and the 30-day forward S&P realized volatility. While our focus is on the R-square in regression results, we also report the mean square error (MSE) and the mean absolute error (MAE) of the residuals from the equations. We also conduct similar regressions with the VIX:

$$Var(R_{t+1}^{S\&P}) = c + \beta * VIX_t + \varepsilon_t$$

In the interests of parsimony, we report the results for the full sample and the 2015–2021 sample (after the last set of changes to the VIX) in Table 1.² The dependent variable in all regressions is the 30-day forward of S&P realized volatility, denoted FVSP, and the key independent variables are the AVE and the VIX.³

Panel A reports results for the full sample. For the regression involving both AVE and VIX, the coefficient estimate of the AVE completely usurps the explanatory power of the VIX. In separate regressions of the FVSP on the AVE and VIX, the estimated coefficients are 1001.87 and 6.45, both significant (note the estimates are multiplied by a factor 10^{-3} for exposition). In the AVE regression the adjusted R-square is more than 4 times larger than the corresponding value for the VIX regression. The very high adjusted R-square of 0.95 suggests the AVE measure is an effective estimator of future realized volatility. By comparison, the CBOE VIX estimator fares relatively poorly – overestimating realized standard deviation substantially and the adjusted R-square is much lower (at 0.20). For robustness we compute the MSE and MAE of the two separate regressions, with the AVE and VIX as independent regressors. The MSE, MAE values for the AVE and VIX are 0.000719, 0.015391 and 0.010479, 0.058885, respectively, thus corroborating the evidence from the R-squares in the regressions. In Panel B of Table 1 we show results for the sub-sample beginning January 2nd 2015 to November 3rd 2021. The results in Panel B mirror the evidence in Panel A. Again, the evidence shows that the AVE outperforms the VIX by substantial margins.

In Fig. 1, we depict out-of-sample forecasts of the FVSP using the AVE and VIX. For this we first regress the FVSP alternately on the AVE and the VIX for the sampling period January 2nd 2015 to November 3rd 2020 and forecast the FVSP for the period November 4th 2020 to November 3rd 2021. From Fig. 1, it becomes apparent that the AVE forecast is simply more accurate than the VIX forecast. All forecast accuracy measures (RMSE, MAE, etc.) confirm this.⁴

4. Conclusion

It is well known that inadequately forecasting a variable that is unobservable, like volatility, can lead to undesirable outcomes (see

² The results for all samples are strikingly similar. The results that are not reported are available from the authors upon request.

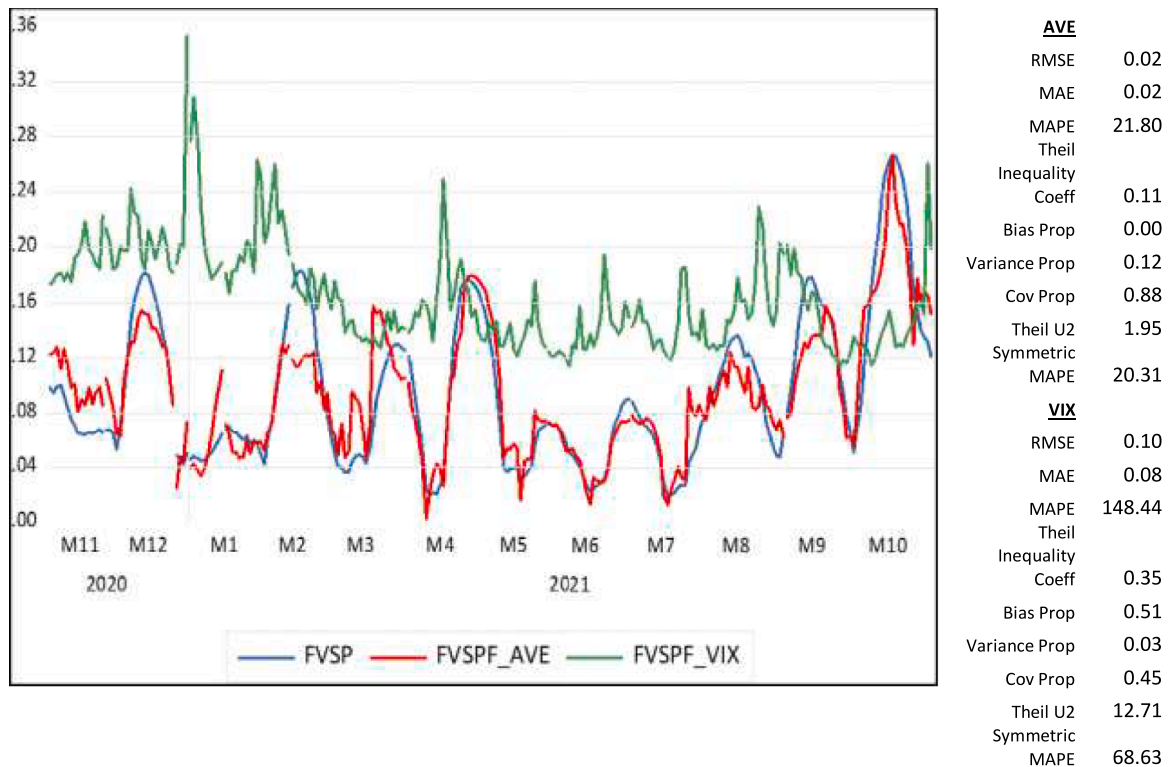
³ Unit root tests suggest the all three series, FVSP, AVE, and VIX are stationary in levels.

⁴ In tests using a sub-sample from 2004 to 2006 we forecast FVSP out-of-sample for the 2008 recessionary period using AVE and VIX as separate explanatory variables. The AVE again outperforms the VIX. These results are available from the authors upon request.

Table 1

Regression of FVSP with AVE and VIX.

Dependent variable:	FVSP					
Panel A: Full Sample 1990 - 2021						
Variable	Coefficient	P-val	Coefficient	P-val	Coefficient	P-val
C	7.13	0.00	2.12	0.00	-16.07	0.00
AVE	1012.95	0.00	1001.87	0.00		
VIX	-0.32	0.00			6.45	0.00
R-squared	0.95		0.95		0.20	
Adjusted R-squared	0.95		0.95		0.20	
Obs	7998.00		7998.00		7998.00	
Panel B: Sample 2015–2021						
Variable	Coefficient	P-val	Coefficient	P-val	Coefficient	P-val
C	21.67	0.00	3.87	0.00	-46.34	0.00
AVE	1019.39	0.00	989.10	0.00		
VIX	-1.23	0.00			10.12	0.00
R-squared	0.96		0.96		0.22	
Adjusted R-squared	0.96		0.96		0.22	
Obs	1723.00		7998.00		7998.00	

Note: Coefficients have been multiplied by a factor 10^3 .**Fig. 1.** Out-of-Sample forecast (11/04/2020 to 11/03/2021) of FVSP with AVE and VIX.

Patton (2011)). There are two major inadequacies in the VIX in this regard. First, it is well known to be extremely asymmetric in returns, and second, it tends to significantly over estimate volatility (e.g., Whaley (2009)). Both inadequacies may be driven by intraday demand pressures in options contracts, perhaps related to jumps in de-risking demand. These characteristics of the VIX (and more generally, of options implied volatility) are consistent enough to suggest improvements. Liu et al., (2022) show that the forecasting ability of VIX can be improved by simply conditioning it on its level. Others employ the VIX for proxying intraday (rather than future) behavior of the underlying asset (see e.g., Lyócsa and Plíhal (2022)).

In this study we provide an alternate estimator of forward volatility, one that is rooted in asymptotic theory. It is an easily implemented metric that is wholly independent of options markets. The estimator we show, greatly outperforms the VIX with a forecasting efficiency of 90%-95%, depending on sampling period. The VIX forecasts correspondingly with an accuracy of about 20%-25%. To inject robustness into our findings we use several goodness-of-fit measures and complement these tests with out-of-sample

forecasts.

The evidence from this study has important implications. One, the VIX, which relies on current prices of options will provide poor and perhaps misleading guidance of future volatility. Two, standard statistical frameworks like the one we employ outperforms the VIX, and three, market participants should regard the VIX as a measure of current sentiment rather than as a volatility-forecasting tool.

CRedit authorship contribution statement

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Declaration of Competing Interest

None.

Data Availability

Data will be made available on request.

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